

Learning target DF2, version 1

Find derivatives using the definition of derivative as a limit.

Suppose that $f(x) = 3x^2 - 5x + 4$. Use the limit definition of the derivative to find $f'(x)$.

PS: The answer is $f'(x) = 6x - 5$.

Learning target DF2, version 2

Find derivatives using the definition of derivative as a limit.

Suppose that $g(w) = 6w^3 - 2w^2 - 9w - 2$. Use the limit definition of the derivative to find $g'(w)$.

Algebra hint: $(w + h)^3 = w^3 + 3w^2h + 3wh^2 + h^3$.

PS: The answer is $g'(w) = 18w^2 - 4w + 9$.

Learning target DF2, version 3

Find derivatives using the definition of derivative as a limit.

Suppose that $p(z) = -2z^3 + 5z^2 - 4z + 4$. Use the limit definition of the derivative to find $p'(z)$.

Algebra hint: $(z+h)^3 = z^3 + 3z^2h + 3zh^2 + h^3$.

PS: The answer is $p'(z) = -6z^2 + 10z - 4$.

Learning target DF2, version 4

Find derivatives using the definition of derivative as a limit.

Suppose that $f(x) = 4x^2 - 3x + 100$. Use the limit definition of the derivative to find $f'(x)$.

PS: The answer is $f'(x) = 8x - 3$.